INCH POUND

MIL-DTL-70507A (AR)
16 March 2010
SUPERSEDING
MIL-P-70507 (AR)
28 December 1984

MILITARY SPECIFICATION

PROJECTILE, 155MM, HE - M864E2
LOADING, ASSEMBLING AND PACKING

Reactivated 16 March 2010 and may be used for new and existing designs and acquisitions.

This specification is approved for use by the U.S. Army Armament Research, Development and Engineering Center (ARDEC), and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the loading, assembling and packing for the projectile, 155MM, HE: M864E2. The M864E2 cargo is the M1162 Pyrotechnic Self-Destruct Fuze (p-SDF) designed to minimize battlefield duds by adding self-destruct capability to the current Dual Purpose Improved Conventional Munitions (DPICM) M42/M46 grenade.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are needed to meet the requirements specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to the commander, U.S. Army ARDEC, ATTN: RDAR-QES-E, Picatinny Arsenal, NJ 07806-5000 or e-mailed to ardec-stdzn@pica.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST online database at https://assist.daps.dla.mil .

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-W-5625	-	Webbing, Textile, Nylon, Tubular
MIL-A-70625	-	Automated Acceptance Inspection Equipment Design, Testing and Approval, of

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-286	-	Propellants, Solid: Sampling, Examination and Testing
MIL-STD-331	-	Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-810	-	Environmental Engineering Considerations and Laboratory Tests
MIL-STD-1168	-	Ammunition Lot Numbering and Ammunition Data Card
MIL-STD-1916	-	DoD Preferred Methods for Acceptance of Product
MIL-STD-3010	-	Test Procedures for Packaging Materials

(Copies of these documents are available online at https://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Bldg. 4D Philadelphia, PA 19111-5094.)

US ARMY DEVELOPMENTAL TEST COMMAND (DTC)

INTERNATIONAL TEST OPERATING PROCEDURES

ITOP 1-2-601	-	Laboratory Vibration Schedule

(These documents may be ordered from the US Army Developmental Test Command, ATTN: Publications, 314 Longs Corner Road, Aberdeen Proving Ground, MD 21005-5005, or online at http://www.dtc.army.mil/publications/topsindex.aspx)

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2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation contract.

DRAWINGS

8841732	-	Leak test fixture
9202990	-	Operating instructions for leak test
9215397	-	Steel shim
9381113	-	Rubber pad
9381111	-	Forward plate
9381117-1	-	Body and ogive assembly for projectile, 155mm, M864
9381130	-	Base burner assembly
9391057	-	Expulsion cup
12934962	-	Charge bag assembly
12934963	-	Charge loading assembly
12944315	-	Expulsion charge assembly
13001888	-	Extended range obturator
13001895	-	Body loading assembly, type PH
13001896	-	Body loading Assembly, type CA
13001900	-	Body fuze assembly, type PH
13001902	-	Body fuze assembly, type CA
13001904	-	Grenade, M46
13001906	-	Grenade, M42
13001907	-	Projectile, 155mm, extended range, DP, M864E2
13001909	-	Tape loop for self destruct fuze
13016773	-	Slider lock for self destruct fuze

(Copies of these drawings may be requested online at pica.drawing.request@conus.army.mil or from US Army ARDEC, RDAR-EIS-PE, Picatinny Arsenal, NJ 07806-5000.)

2.3 Non-Government publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS

TITLE 49 - Transportation, Parts 100-199

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders should specify, “49 CFR 100-199 (latest revision)”.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

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ASTM E 18	-	Standard Test Methods for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials
ASTME E 1742	-	Standard Practice for Radiographic Examination

(Copies of these documents are available from www.astm.org or ASTM International 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Conformance inspection. A sample shall be subjected to conformance testing in accordance with 4.3.

3.3 Components and assemblies. Materials, parts and assemblies shall comply with drawings cited in 4.3.2.1 through 4.3.2.18.

3.4 Projectile. The projectile assembly and all components thereof shall comply with all requirements specified on drawing 13001907.

3.5 Ballistic functioning.

3.5.1 Functioning (not including safety phase). All the projectile hardware shall remain intact until the initiation of the time fuze at a predetermined time after firing. The delivery of the cargo to the target area shall also include the successful expulsion of the entire cargo of grenades.

3.5.2 Functioning. There shall be no premature burst or evidence thereof and there shall be no separation of metal parts either in the bore of the weapon or in flight (prior to initiation of the time fuze) in any of the ballistic functioning tests. This requirement shall also apply to those rounds to be fired in the safety phase.

3.5.3 Grenade performance. The functional reliability shall be not less than ninety-nine (99) percent resulting in an Unexploded Ordnance (UXO) rate of not greater than one (1) percent.

3.5.4 Thermal shock. The projectile shall function after exposure to alternatively high and low temperature extremes.

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3.5.5 Transportation vibration. The projectile shall show no degradation in safety or functionality after exposure to secured cargo vibration.

3.6 Radiographic examination of projectiles. Prior to forwarding the projectiles to the proving grounds for ballistic testing, the projectiles shall be subjected to radiographic examination in accordance with (13001907) and be in accordance with ASTME E 1742. The projectiles shall be examined for the following:

a. Gap between adapter and base (see f).
b. Gap between adapter and grenades (see f).
c. Gap between grenades (see f).
d. Any missing or improperly positioned component or assembly normally discernible on the radiograph.
e. Any missing or misassembled base burner component, including ignitor cup loading assembly facing the wrong direction and foreign matter present.
f. Gaps in excess of 0.015 inches shall be considered rejects.

3.7 Ammunition lot numbering. Ammunition lot numbers shall be assigned in accordance with MIL-STD-1168.

3.8 Workmanship. Molded component parts shall be free from splits, cold flow, shrinkage, flash, gouges, cold shuts, inclusion, porosity, or any similar characteristic. All parts shall be free of burrs, sharp edges, cracks, dirt, grease, rust, and other foreign matter. Care shall be exercised at all times so that the rotating band of each projectile will not be nicked or burred. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous; however, a few light scratches not exposing base material are permitted. All required marking and stamping shall be neat and sharply defined.

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4. VERIFICATION

TABLE I. Requirement and verification cross reference

Methods of verification	Classes of verification
1. Analysis 2. Demonstration 3. Examination 4. Test	A. First article B. Conformance inspection

Section 3 Requirements	Description	Verification method				Verification class		Section 4 Verification method
		1	2	3	4	A	B	
3.1	First article			x	x	x		4.2
3.2	Conformance inspection			x	x		x	4.3
3.3	Components and assemblies			x	x	x	x	4.3.2.1-4.3.2.18
3.4	Projectile			x	x	x	x	4.3.2.18
3.5	Ballistic functioning				x	x	x	4.4.10
3.5.1	Functioning (non-safety)				x	x	x	4.4.10
3.5.2	Functioning				x	x	x	4.4.10
3.5.3	Grenade performance				x	x	x	4.4.10, 4.4.10.2
3.5.4	Thermal shock				x	x		4.4.10.5
3.5.5	Transportation vibration				x	x		4.4.10.6
3.6	Radiographic examination			x		x	x	4.4.16
3.7	Ammunition lot numbering			x		x	x	4.3.2.18
3.8	Workmanship			x		x	x	4.3.2.1 – 4.3.2.18

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

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4.2 First article. When required, a first article sample shall be evaluated in accordance with 4.2.1.

4.2.1 First article quantity. Samples shall be tested in accordance with Table II.

TABLE II. First article inspection

Examination or test	Conformance criteria	Requirement paragraph	Inspection method reference
Grenade assembly	16	3.3	4.3.2.4
Steel shim	25	3.3	4.3.2.5
Rubber pad	10	3.3	4.3.2.6
Forward plate	10	3.3	4.3.2.7
Expulsion cup	16	3.3	4.3.2.8
Charge bag assembly	16	3.3	4.3.2.9
Charge loading assembly	16	3.3	4.3.2.10
Expulsion charge assembly	16	3.3	4.3.2.11
Extended range obturator	5	3.3	4.3.2.12
Base burner assembly	16	3.3	4.3.2.13
Projectile metal part, prior to loading	16	3.3	4.3.2.14
Projectile examination for defects	21 projectiles	3.4	4.3.2.15-4.3.2.18
Ballistic functioning	21 projectiles, 1/	3.5	4.4.10
Transportation vibration	12 projectiles, 1/	3.5.5	4.4.10.6
Thermal shock	12 projectiles, 1/	3.5.4	4.4.10.5
Radiographic inspection	2/	3.6	4.4.16

NOTES:
1/ See figure 1 (see 6.19).
2/ The 21 projectiles that are used for radiographic inspection are the same that are ballistically functioned.

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4.2.2 First article rejection. If any item of the sample fails to comply with the first article requirements, the sample shall be rejected. The first article ballistic functioning test shall be rejected if the sample experiences any of the following ballistic rejection criteria:

a. Any premature burst occurs.
b. Any separation of metal parts occurs prior to initiation of the time fuze.
c. Any detonation of projectile cargo occurs prior to initiation of the time fuze.
d. Unexploded ordnance on ground exceeds 1%.

4.3 Conformance inspection.

4.3.1 Lot formation. Lot formation shall be in accordance with the lot formation requirement of MIL-STD-1916, Acceptance by tables. In addition, inspection lots of finished projectiles shall contain:

a. Loaded projectiles of one weight zone.
b. Expulsion cups from one lot interfix number from one manufacturer.
c. Fuzes, M1162 p-SDF, from one lot interfix number from one manufacturer.
d. Slider lock feature from one lot interfix number from one manufacturer.
e. Tape loop from one lot interfix number from one manufacturer.
f. Washer from one lot interfix number from one manufacturer.

4.3.2 Inspection requirements by classification of characteristics.

a. For the conformance inspection paragraphs 4.3.2.1 to 4.3.2.18, the definitions of critical, major and minor are provided in paragraph “Definitions” of MIL-STD-1916.
b. The acceptance criteria for sampling inspection shall be in accordance with the levels provided in conformance examination/test paragraph and MIL-STD-1916.
c. Alternative conformance acceptance. Alternate conformance procedures may be proposed (see paragraph “Acceptance by contractor proposed provisions” of MIL-STD-1916).
d. Automated Acceptance Inspection Equipment (AAIE) shall be designed, qualified and calibrated in accordance with MIL-A-70625. AAIE shall automatically determine conformance of a characteristics without operator analysis or intervention.
e. Inspection sampling requirements for critical, major and minor characteristics are defined in MIL-STD-1916. Unless specified otherwise, Inspection Level VII in addition to 100% inspection shall be used for all characteristics defined as Critical, Inspection Level IV shall be used for all characteristics defined as Majors and Inspection Level II for all minor characteristics.

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4.3.2.1	Body loading assembly (after M223 fuze removal).			Drawing number 13001895, 13001896
Classification	Examination or test	Conformance criteria	Requirement paragraph	Next higher assembly / Inspection method reference
Critical	None defined			
Major				
101	Lead cup assembly damaged, split, cracked, missing or improperly assembled	100%	3.3	AIE
102	Lead above flush or below max depth	Level IV	3.3	AIE
103	Stud loose or free to rotate, after fuze removal	Level IV	3.3	AIE
104	Depth of drill	Level IV	3.3	AIE
105	Any body cracked, dented or distorted after fuze removal	Level IV	3.3	Visual
Minor				
201	Evidence of poor workmanship	Level II	3.8	Visual

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4.3.2.2	Tape loop for fuze.			Drawing number 13001909
Classification	Examination or test	Conformance criteria	Requirement paragraph	Next higher assembly Inspection method reference
Critical	None defined			
Major				
101	Residual shrinkage test	1/	3.3	4.4.11
102	Tape overlap welded improperly	Level IV	3.3	Visual
103	Maximum overlap distance between ends	Level IV	3.3	Visual
104	Thickness	Level IV	3.3	Gage
105	Width	Level IV	3.3	Gage
106	Length	Level IV	3.3	Gage
Minor				
201	Evidence of poor workmanship	Level II	3.8	Visual

NOTES:
1/ If tape loop assembled at the fuze level from fuze manufacturer with tape loop, then this process does not need to be repeated at the LAP level.

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4.3.2.3	Slide lock for fuze.			Drawing number 13016773
Classification	Examination or test	Conformance criteria	Requirement paragraph	Next higher assembly Inspection method reference
Critical	None defined			
Major				
101	Thickness prior to forming	Level IV	3.3	AIE
102	Slot profile 2 places prior to forming	Level IV	3.3	AIE
103	Outside profile prior to forming	Level IV	3.3	AIE
104	Angle of slide lock prior to forming	Level IV	3.3	AIE
105	Inside radius	Level IV	3.3	AIE
106	Distance to bottom of slide lock	Level IV	3.3	AIE
107	Radius of slide lock tab (2 places)	Level IV	3.3	AIE
108	Radius of slotted tab (2 places)	Level IV	3.3	AIE
109	Total width	Level IV	3.3	AIE
110	Distance across tabs at bottom	Level IV	3.3	AIE
111	Distance to outside of slide lock	Level IV	3.3	AIE
112	Distance to end of slide lock	Level IV	3.3	AIE
Minor				
201	Evidence of poor workmanship	Level II	3.8	Visual

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4.3.2.4	Grenade assembly (prior to loading into projectile).			Drawing number 13001904, 13001906
Classification	Examination or test	Conformance criteria	Requirement paragraph	Next higher assembly 13001907 Inspection method reference
Critical				
1	Slide protruding excessively from fuze indicating armed or partially armed condition	100%, Level VII, 3/	3.3	AAIE
2	Slider lock feature missing or not seated at correct height	100%, Level VII	3.3	AAIE
3	Trigger missing or out of position	100%, Level VII, 3/	3.3	AAIE
4	Safety pin missing	100%, Level VII	3.3	AAIE
5	Safety clip missing	100%, Level VII, 2/	3.3	AAIE
Major				
101	Tape loop incorrectly contained by slide lock.	Level IV	3.3	Visual
102	Arming screw clinch missing, incomplete, or not properly centered	Level IV	3.3	Visual
103	Tape loop/washers missing or incorrect	100%	3.3	Visual
104	Fuze housing cracked or deformed	Level IV	3.3	Visual
105	Stud clinch missing	Level IV	3.3	Visual
106	Depth of recapped stud	Level IV	3.3	AIE
107	Unthreading torque test of clinched arming screw	1/	3.3	4.4.15.2
108	Arming test of fuze slide	1/	3.3	4.4.15.3
109	Tape loop pull test	1/	3.3	4.4.15.4
110	Pull test of clinched studs	1/	3.3	4.4.15.5
111	Tape loop missing or incorrect	Level IV	3.3	Visual

NOTES:
1/ Two samples per 1800 grenade assemblies produced on each machine. Accept on 0, reject on 1failure.
2/ Safety clip shall be removed immediately prior to grenade assembly insertion into projectile.
3/ Examination shall be performed prior to slide lock being fitted onto fuze grenade assembly.

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4.3.2.4	Grenade assembly (prior to loading into projectile).			Drawing number 13001904, 13001906
Classification	Examination or test	Conformance criteria	Requirement paragraph	Next higher assembly 13001907 Inspection method reference
112	Arming screw full thread visible	Level IV	3.3	Visual
113	Arming screw free rotation	Level IV	3.3	Manual
114	Slider housing clearance	Level IV	3.3	AIE
115	Slider lock disturbed	Level IV	3.3	Visual
Minor	None defined			

NOTES:
1/ Two samples per 1800 grenade assemblies produced on each machine. Accept on 0, reject on 1failure.
2/ Safety clip shall be removed immediately prior to grenade assembly insertion into projectile.
3/ Examination shall be performed prior to slide lock being fitted onto fuze grenade assembly.

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4.3.2.5	Steel shim.			Drawing number 9215397
Classification	Examination or test	Conformance criteria	Requirement paragraph	Next higher assembly 13001907 Inspection method reference
Critical	None defined			
Major				
101	Hardness	Level IV	3.3	4.4.1
102	Diameter, min	Level IV	3.3	AIE
103	Thickness, min.	Level IV	3.3	AIE
Minor				
201	Evidence of poor workmanship	Level II	3.8	Visual

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4.3.2.6	Rubber pad.			Drawing number 9381113
Classification	Examination or test	Conformance criteria	Requirement paragraph	Next higher assembly 13001907 Inspection method reference
Critical None defined				
Major 101	Diameter	Level IV	3.3	AIE
Minor 201	Thickness	Level II	3.3	AIE

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4.3.2.7	Forward plate.			Drawing number 9381111
Classification	Examination or test	Conformance criteria	Requirement paragraph	Next higher assembly 13001907 Inspection method reference
Critical	None defined			
Major				
101	Aft outside diameter	Level IV	3.3	AIE
102	Diameter of O-ring seat	Level IV	3.3	AIE
103	True position of O-ring seat	Level IV	3.3	AIE
104	Thickness	Level IV	3.3	AIE
105	Width of O-ring seat	Level IV	3.3	AIE
106	Location of basic diameter on taper	Level IV	3.3	AIE
107	Distance from rear face to forward face of O-ring seat	Level IV	3.3	AIE
108	O-ring not greased prior to assembling to forward plate	Level IV	3.3	Visual
Minor				
201	Evidence of poor workmanship	Level II	3.8	Visual

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4.3.2.8	Expulsion cup.			Drawing number 9391057
Classification	Examination or test	Conformance criteria	Requirement paragraph	Next higher assembly 13001907 Inspection method reference
Critical	None defined			
Major				
101	Pitch diameter of thread	Level IV	3.3	AIE
102	Major diameter of thread	Level IV	3.3	AIE
103	True position of largest inside diameter with pitch diameter of threads	Level IV	3.3	AIE
104	True position of smallest inside diameter with pitch diameter of thread	Level IV	3.3	AIE
105	Minimum inside diameter	Level IV	3.3	AIE
106	True position of largest outside diameter with pitch diameter of thread	Level IV	3.3	AIE
107	Length of threaded area	Level IV	3.3	AIE
108	Thickness of bottom	Level IV	3.3	AIE
109	Thickness of sidewall	Level IV	3.3	AIE
110	Overall length	Level IV	3.3	AIE
111	Depth to start of angle	Level IV	3.3	AIE
112	Presence of two slots	Level IV	3.3	Visual
113	Largest outside diameter	Level IV	3.3	AIE
114	Perpendicularity of largest outside diameter	Level IV	3.3	AIE
115	Presence of angle between large and small inside diameters	Level IV	3.3	AIE
116	Expulsion hardness test	Level IV	3.3	4.4.2
Minor				
201	Evidence of poor workmanship	Level II	3.8	Visual

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4.3.2.9	Charge bag assembly (required for new production only.			Drawing number 12934962
Classification	Examination or test	Conformance criteria	Requirement paragraph	Next higher assembly 12934963 Inspection method reference
Critical	None defined			
Major				
101	Length of bag	Level IV	3.3	AIE
102	Distance to end of heat seal max.	Level IV	3.3	AIE
103	Distance to side heat seal	Level IV	3.3	AIE
104	Minimum width of both heat seals over entire length	Level IV	3.3	AIE
105	Heat seals incomplete (2 places)	Level IV	3.3	Visual
106	Strength of bag charge assembly heat seal	1/	3.3	4.4.3
107	Location of end stitching	Level IV	3.3	AIE
108	Distance to side stitching	Level IV	3.3	AIE
109	Location of eyelet	Level IV	3.3	AIE
Minor				
201	Stitching incomplete (2 places)	Level II	3.3	Visual
202	Fold width eyelet end	Level II	3.3	AIE
203	Fold misalignment over max.	Level II	3.3	AIE
204	Thickness of assembled eyelet	Level II	3.3	AIE
205	Evidence of poor workmanship	Level II	3.8	Visual

NOTES:
1/ Sample 20 per lot, accept on 0, reject on 1 failure.

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4.3.2.10	Charge loading assembly.			Drawing number 12934963
Classification	Examination or test	Conformance criteria	Requirement paragraph	Next higher assembly 12944315 Inspection method reference
Critical	None defined			
Major				
101	Pull wire of bag loose with respect to cover	Level IV, 1/	3.3	Visual
102	Cover damaged	Level IV, 1/	3.3	Visual
103	Clinch of stud improper	Level IV	3.3	Visual
104	Bag damaged	Level IV	3.3	Visual
Minor				
201	Evidence of poor workmanship	Level II	3.8	Visual

NOTES:
1/ 101 and 102 required for re-used charge loading assembly.

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4.3.2.11	Expulsion charge assembly.			Drawing number 12944315
Classification	Examination or test	Conformance criteria	Requirement paragraph	Next higher assembly 13001907 Inspection method reference
Critical	None defined			
Major				
101	Leak test of expulsion charge	1/, 2/	3.3	4.4.4
102	Weight of expulsion charge incorrect	Level IV 2/	3.3	AIE
103	Bag seal incomplete (3 places)	Level IV	3.3	Visual
104	Moisture and total volatiles content of propellant	1/, 2/	3.3	4.4.5
105	Bag damaged	Level IV	3.3	Visual
Minor				
201	Evidence of poor workmanship	Level II	3.8	Visual

NOTES:
1/ Sample in accordance with applicable test method paragraph.
2/ 101, 102 and 104 required for new production only.

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4.3.2.12	Extended range obturator.			Drawing number 13001888
Classification	Examination or test	Conformance criteria	Requirement paragraph	Next higher assembly 13001907 Inspection method reference
Critical	None defined			
Major				
101	Outside diameter	Level IV	3.3	AIE
102	Wall thickness	Level IV	3.3	AIE
103	Wall thickness at grooves	Level IV	3.3	AIE
104	Inside diameter of forward end	Level IV	3.3	AIE
105	Width of forward ID	1/	3.3	4.4.6
106	Length of obturator	Level IV	3.3	AIE
107	Basic forward internal angle	1/	3.3	4.4.6
108	Length of forward outside angle	1/	3.3	4.4.6
109	Breaking load test	2/	3.3	4.4.7
110	Flash on front face	Level IV	3.3	Visual
111	Flashing on rear face maximum	Level IV	3.3	Visual
112	Gates and knits are to be located away from the grooves	Level IV	3.3	Visual
Minor				
201	Basic aft outside angle	1/	3.3	4.4.6
202	Basic aft inside angle	1/	3.3	4.4.6
203	Length to rear inside angle	1/	3.3	4.4.6
204	Length from forward face to aft step	1/	3.3	4.4.6
205	Groove width	Level II	3.3	AIE
206	True position of grooves	Level II	3.3	AIE
207	Location of groove sets	Level II	3.3	AIE
208	Molding defects	Level II	3.3	Visual
209	Color	Level II	3.3	Visual
210	Workmanship	Level II	3.8	Visual

NOTES:
1/ Two obturators per mold per shift shall be sampled.
2/ Five obturators per mold per shift shall be sampled.

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4.3.2.13	Base burner assembly.			Drawing number 9381130
Classification	Examination or test	Conformance criteria	Requirement paragraph	Next higher assembly 13001907 Inspection method reference
Critical	None Defined			
Major				
101	Base threads not cleaned or damaged	Level IV	3.3	Visual
102	Paint damaged or missing	Level IV	3.3	Visual
103	Excessive gap between base and base closure (360º)	Level IV	3.3	AIE
Minor				
201	Marking missing, incorrect or unidentifiable including propellant lot and batch identification	Level II	3.3	Visual
202	Evidence of poor workmanship	Level II	3.8	Visual

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4.3.2.14	Body and ogive assembly.			Drawing number 9381117-1
Classification	Examination or test	Conformance criteria	Requirement paragraph	Next higher assembly 13001907 Inspection method reference
Critical	None defined			
Major				
101	Threads damaged or improperly cleaned at aft end	Level IV	3.3	Visual
102	Threads damaged or improperly cleaned on lifting plug end	Level IV	3.3	Visual
103	Internal lubricant applied improperly	Level IV	3.3	Visual
104	Paint missing or damaged	Level IV	3.3	Visual
105	Gap between Body and Ogive	Level IV	3.3	AIE
106	Corroded or damaged Rotating Band	Level IV	3.3	Visual
Minor				
201	Foreign material on interior surface of body	Level II	3.3	Visual
202	Evidence of poor workmanship	Level II	3.8	Visual

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4.3.2.15	Projectile (after each layer of grenades are inserted and grenade safety pin and clip are removed).			Drawing number 13001907
Classification	Examination or test	Conformance criteria	Requirement paragraph	Next higher assembly Inspection method reference
Critical				
1	Slide protruding excessively from fuze indicating armed or partially armed condition	100%, Level VII	3.3	AAIE
2	Layer 7-9 grenade type incorrect (yellow marking not present)	100%, Level VII	3.3	AAIE
3	Slider lock feature missing or not seated at correct height	100%, Level VII	3.3	AAIE
4	Any grenade missing	100%, Level VII	3.3	AAIE
5	Any sleeve, spacer, or spline missing	100%, Level VII	3.3	AAIE
Major				
101	Layer 1-6 grenade type incorrect	Level IV	3.3	Visual
102	Any grenade safety features (safety pin and clip) not removed	100%, 1/	3.3	Visual
103	Any sleeve, spacer, or spline improperly positioned or incorrect length	Level IV	3.3	Visual
Minor	None defined			

NOTES:
1/ Visually inspect each row of grenades for presence of safety clip/pin before adding next layer of grenades.

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4.3.2.16	Projectile (after assembly of adapters).			Drawing number 13001907
Classification	Examination or test	Conformance criteria	Requirement paragraph	Next higher assembly Inspection method reference
Critical				
1	Adapter or adapter insert missing	100%, Level VII	3.3	AAIE
2	Any sleeve, spacer or spline missing	100%, Level VII	3.3	AAIE
Major				
101	Any sleeve, spacer or spline improperly positioned or incorrect length	Level IV	3.3	Visual
102	Adapter or adapter insert improperly assembled	Level IV	3.3	Visual
Minor	None defined			

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4.3.2.17	Projectile (prior to assembling base burner assembly after o-ring has been applied).			Drawing number 13001907
Classification	Examination or test	Conformance criteria	Requirement paragraph	Next higher assembly Inspection method reference
Critical	None defined			
Major				
101	Depth between aft face of body and cargo (shimmed as applicable) at specified load	100%	3.3	4.4.8
102	O-ring missing or wrong type	Level IV	3.3	Visual
103	O-ring not greased	Level IV	3.3	Visual
104	Thread adhesive missing	100%	3.3	Visual
105	Threads damaged on base burner assembly or projectile body	Level IV	3.3	Visual
106	Foreign material on threads	Level IV	3.3	Visual
Minor				
201	Sealing compound not present between body and base, 360 degrees	Level II	3.3	Visual
202	Evidence of poor workmanship	Level II	3.8	Visual

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4.3.2.18	Projectile (after assembly of base burner assembly).			Drawing number 13001907
Classification	Examination or test	Conformance criteria	Requirement paragraph	Next higher assembly Inspection method reference
Critical				
1	Excessive gap between projectile body and base burner	100%, Level VII	3.3	4.4.9
2	Ballistic functioning, safety	3 projectiles, 2/	3.5	4.4.10.2
Major				
101	Disassembly torque test of base burner assembly	100%	3.3	4.4.12
102	Disassembly torque of lifting plug assembly	100%	3.3	4.4.13
103	Air pressure test of projectile	100%,	3.3	4.4.14
104	Radiographic inspection of projectiles	100%, 1/	3.6	4.4.16
105	Diameter of bourrelet, max (applicable after paint on projectile has been retouched)	100%	3.3	AIE
106	Prick missing, incorrect or unidentifiable	Level IV	3.3	Visual
107	Diamond marking missing, incorrect or unidentifiable	Level IV	3.3	Visual
108	Paint markings missing, incorrect or unidentifiable	Level IV	3.3	Visual
109	Sealant around weather seal to base closure interface missing or not 360 degrees	100%	3.3	Visual
110	Ballistic functioning	15 projectiles, 2/	3.5	4.4.10.1
111	Ammunition lot number	100%	3.7	Visual
112	Rotating band damaged	Level II, 3/	3.3	Visual
113	Extended range obturator missing	Level II	3.8	Visual
114	Transportation vibration	12 projectiles, 2/	3.5.5	4.4.10.6
115	Thermal shock	12 projectiles, 2/	3.5.4	4.4.10.5
Minor				
201	Missing or disturbed flexible rotating band cover	Level II	3.8	Visual
202	Evidence of poor workmanship	Level II	3.8	Visual

NOTES:
1/ Ballistic samples only.
2/ See figure 2 (see 6.19).
3/ See 6.14.8

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4.4 Methods of inspection.

4.4.1 Hardness test of shim. This test shall be performed in accordance with ASTM E18 using the standard penetractor for the specified hardness scale. Three readings shall be taken approximately 120 degrees apart. If a reading fails the requirement of drawing 9215397, that shim shall be classed defective.

4.4.2 Hardness test of expulsion cup. The cups selected shall be subjected to the hardness test in accordance with the method and procedures of ASTM E18. The load and penetractor shall be in accordance with the hardness scale specified on the applicable drawing. Three readings shall be taken on each cup on the outside surfaces of the closed end and approximately ¼ inch from the edge and approximately 120 degrees apart. A suitable fixture shall be utilized to provide support under the surface being tested. If the average value of the three readings for any cup is less than the value specified in the applicable drawing, that cup shall have failed the requirement. If any sample expulsion cup is found to be in excess of the requirements of drawing 9391057, the inspection lot it represents shall be rejected.

4.4.3 Strength of bag charge assembly heat seal. This test shall be performed in accordance with MIL-STD-3010 as modified herein. One specimen, one inch wide, shall be cut perpendicular to the line of a seal of each sample bag. Location of the sample shall be so that both the end seam and the long seam are sampled. One leg of the specimen shall be clamped so that the specimen hangs freely. The required load (2.5 lbs.) shall be applied to the free end such as to avoid impact loading the seam. The load shall be allowed to act for a minimum of one minute after which the seam shall be examined for peel-back of the heat seal. The test sample shall be conditioned prior to testing at a temperature of 70°F to 80°F and shall be tested under those conditions unless otherwise specified. If any sample heat seal is found to be in excess of the requirements of drawing 12934962, the inspection lot it represents shall be rejected.

4.4.4 Leak test of expulsion charge assembly. A sample of 5 expulsion charge assemblies from each hour of production per sealing machine shall be subjected to the vacuum leak test specified on drawing 12944315 using equipment and procedures shown on drawings 8841732 and 9202990 (except test instruction 4, see 4.4.4.c for reject criteria). If any sample assembly fails the leak test (see 4.4.4.c), the quantity represented by that sample shall be set aside and retested per the following procedures:

a. Visually inspect all bags for completeness of heat seals (top, side and bottom). Bags with damaged or incomplete seals shall be classed defective and removed from production.

b. From the remaining assemblies, ten (10) shall be selected and subjected to the leak test, accept on 0 defects, reject on 1 defect.

c. For the purpose of this test, a reject shall an assembly that leaks to the extent that the propellant becomes wet. To determine the condition of the propellant, those assemblies that exhibit darkening (an indication of wet propellant) of the propellant shall be cut open and the propellant shall be visually and manually inspected for moisture. Only those assemblies

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containing moisture shall be considered defective. Assemblies that do not contain moisture, for lots that have passed leakage testing, shall have the propellant salvaged. There shall be no mixing of lots of salvaged propellant. Salvaged propellant shall be tested in accordance with 4.4.5.

4.4.5 Moisture and total volatiles content of propellant. The moisture and total volatiles content of each lot of propellant or portion thereof shall be determined for each shift (not to exceed 10 hours) of production. This determination shall be accomplished prior to the anticipated use of the lot or portion thereof, with a maximum of eighteen (18) hours lead time. Sampling of the propellant shall be in accordance with MIL-STD-286, Method 101.1. If the moisture and total volatile content fails to comply with the requirements on drawing 12944315, the propellant lot shall be rejected until corrective action has been taken and verified by repeating the moisture and total volatile content determination.

4.4.6 Sectioned obturator. Two obturators per mold per shift plus the last two obturators per mold from a continuous production run shall be sampled. Sample shall be verified through AIE. If any sample obturator is found to be in excess of requirements on drawing 13001888, the inspection lot it represents shall be rejected.

4.4.7 Breaking load test. Five obturators per mold per shift from a continuous production run shall be tested. A shift shall not exceed 10 hours of production. The obturator shall be tested for breaking load using an approved test fixture. The test fixture shall consist of two 76.2 mm (3 inch) radius semicircular steel locating discs machined to the internal profile of the obturator. The test speed shall be 50.8 mm (2.00 inch) per minute.

4.4.8 Depth to adapter shims. This measurement shall be performed using automatic inspection equipment prior to assembling base burner assembly (after o-ring has been applied). The design of this equipment shall be ‘fail-safe’ to the extent that the projectile cannot be released for further processing until the proper numbers of shims have been inserted. Verification that the proper numbers of shims have been inserted shall be accomplished by re-inspection of this depth.

4.4.9 Inspection of gap between projectile body and base burner. The base to body joint shall be inspected for gaps using an approved feeler gage as specified on drawing 13001907 after assembly of base burner assembly to projectile. The feeler gage shall be cleaned with a clean dry rag immediately after each inspection such that the gage is free of any base/body gap sealant. No solvent or other substance other than a clean dry rag shall be used to clean the gage. The gap between the projectile body and base shall be inspected within 15 minutes after the sealant is applied to the shoulder of the base.

4.4.10 Ballistic functioning.

4.4.10.1 Ballistic functioning (not including safety phase). Care shall be exercised at all times so that the rotating band of each projectile will not be nicked or burred. The round shall be fitted with a M762A1 fuze. The M762A1 fuze setting and quadrant elevation (QE) of the weapon shall be adjusted to attempt to achieve ejection over a prepared impact area at the target height

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specified below. During any test phase the fuze setting and QE shall remain constant. The maximum ejection height shall be 1,800 feet, for any zone. A photograph shall be taken of the fuze setting for each round immediately prior to use and identified with the round number. Projectiles shall be fired from an M198 cannon. Gun tubes used for this test shall not have less than 25 percent remaining tube life (for wear) based on current gun tube condemnation criteria. The ballistic sample shall be rejected if UXO on proving ground does not meet the requirements of 3.5.3.

4.4.10.2 Ballistic functioning (in the safety phase). Projectiles shall be fired with the same method as 4.4.10.1 with the exception of the propellant charge. The charge at the proving ground shall be adjusted to reach approximately 60,300 psi with the projectile conditioned to the required temperatures. The ballistic safety test shall cause reject of the lot if projectile hardware does not remain intact until initiation of artillery projectile fuze.

4.4.10.3 Exclusions. If, for any reason, the proving ground considers that test conditions have affected all or part of the test results, the proving ground shall declare the affected results invalid and perform new tests on replacement projectiles as required. The following are a list of known exclusions for guidance to the evaluator:

a. Low bursts. If any test projectile functions below the minimum burst height (900 feet), and the result of that firing would cause rejection based solely on grenade related defects, a replacement round shall be fired.

b. Lost grenades. If any test projectile has eight (8) or more lost grenades, a replacement round is to be fired. The grenades that were found shall be scored and added to the overall population. If less than eight (8) grenades lost, a replacement round shall not be fired and what is found shall be scored and added to the overall population.

c. Missing fuzes. High explosive grenades which are recovered without fuzes shall be counted as UXO.

4.4.10.4 Grenade air detonation. Grenade air detonations shall be recorded for informational purposes.

4.4.10.5 Thermal shock. Testing for thermal shock shall be done in accordance with MIL-STD-810, Temperature Shock. The projectile shall be placed inside an environmental chamber set at -50°F for initiation of the test sequence. At completion of the first 4-hour cycle, the projectile shall be transferred to a high temperature chamber set at 145°F for a 4-hour cycle. This process shall be repeated until six cycles (48 hours of exposure) are completed. The projectile will then be stored under ambient condition for 48 hours. If the projectile fails to function per 4.4.10.1, test is rejected.

4.4.10.6 Transportation vibration. Testing for transportation vibration shall be done in accordance with ITOP 1-2-601, Appendix B, Table T-I. If the projectile fails to function per 4.4.10.1, test is rejected.

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4.4.11 Tape loop residual shrinkage. The tape shall be tested in accordance with MIL-W-5625, paragraph 3.6. If any tape fails to comply with the requirement specified on the applicable drawing, the lot shall be rejected.

4.4.12 Disassembly torque test of base burner assembly. The projectile with base assembled, shall be subjected to the disassembly torque specified on the applicable drawing. Any unit that fails to withstand the disassembly torque shall be classed defective.

4.4.13 Disassembly torque of lifting plug assembly. The projectile, with lifting plug assembled, shall be subjected to the disassembly torque specified on the applicable drawing. Any unit that fails to withstand the disassembly torque shall be classed defective.

4.4.14 Air pressure test of projectile. The assembled projectile, with lifting plug, shall be placed in one or more chambers (the base to body, base to base closure, base closure seal, body to ogive and ogive to lifting plug joint must be tested), and a predetermined volume of air shall be delivered to produce the specified pressure between the inside of the chamber and the outside of the projectile. After initiation of test, the pressure must achieve the required level and maintain that level for the specified time interval (see drawing 13001907). Failure to meet either criterion shall be cause for rejection and that projectile shall be suitably identified and removed from the lot to prevent mixing with accepted projectiles. All equipment used in this test shall be capable of performing the test accurately and consistently. Equipment shall be calibrated at the start of each shift and at intervals of not more than one half-shift (not to exceed 5 hours) of continuous operation. It shall also be calibrated prior to restarting the operation after a stop in processing of 3 hours or more. If it is found that the gage is out of calibration such that the test pressure was actually lower than that specified on the drawing, projectiles tested since the last satisfactory calibration shall be retested after the gage error has been corrected.

4.4.15 Grenade assembly.

4.4.15.1 Preparation of sample items. The samples to be subjected to this inspection shall consist of a fuze, a tape stiffener assembly, and an empty body assembly prepared from lots used in production of the grenade recapitalization lot. They shall be prepared at the specified times from each assembly machine used during lot production. The samples shall be suitably marked to prevent mixing with grenades not intended to be samples, and to allow identification of the particular assembly machine on which the unit was produced.

4.4.15.2 Unthreading torque test. The sample shall be mounted in the approved fixture. The safety features shall be removed. With the fixture preventing the fuze slide from arming, the torque required to rotate the arming screw out of the slide shall be measured. If the torque exceeds the amount specified on drawings 13001904 and 13001906, the sample shall be classed defective.

4.4.15.3 Fuze slide arming test. After completion of the unthreading torque test, while maintaining the arming screw free of the slide, the fixture device for preventing slide movement shall then be withdrawn permitting the slide to move to the armed position. After observation for

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compliance with the arming requirements and while still maintaining the arming screw free of the slide, the slide shall be returned to the unarmed position (which requires overriding any lock that holds the slide in the armed position). The safety pins and safety clip shall be properly reinserted in the fuze. Grenade fuzes failing to comply with the specified arming indication shall be classed defective.

4.4.15.4 Tape loop pull test. After completion of the arming test, the same sample shall be mounted in an approved fixture and subjected to a tape loop pull test. A 130lb force shall be exerted on sample. If tape loop fails to remain attached, the sample shall be classed defective.

4.4.15.5 Pull test of clinched studs of body fuze assembly. After completion of the tape loop pull test, the same fuzes shall be subjected to the fuze pull-off test. For this test the tape loop shall be removed from the fuze without damaging the arming screw threads. All shipping safeties, if present, shall be removed. The arming screw shall be unthreaded to the armed position. A fixture shall engage the arming screw and pull the grenade body on the other end, in the direction shown on the drawing, in an attempt to separate the fuze from the grenade. The p-SDF grenade assembly shall withstand the required minimum pull force. After application of the tensile force, the gap between the entire interface of the housing and the cover and between the entire interface of the cover and the grenade shall be measured. The sum of both measurements shall not exceed 0.005 inch.

4.4.15.6 Disposition. If any body assembly fails to comply with any of these tests, the production quantity represented by that sample shall be rejected.

4.4.16 Radiographic inspection of projectiles. The projectile assemblies shall be radiographed prior to forwarding to the proving ground. Prior to assembly of the base burner assembly, the location of the spline shall be indicated by marking the outside of the projectile. In all exposures the radiographic beam shall be focused along the center axis of the projectile and parallel to the face of the forward plate, with the spline 90 degrees to the right in the direction from the source to the projectile. Radiographic exposure shall as a minimum, include all of the base burner assembly, to and including the entire forward plate. If necessary, more than one radiograph shall be taken in this view if required to avoid detrimental parallax effects and to observe the conditions specified in 3.6. In the event that the radiograph(s) taken in this view prevent a suitable examination, the projectile shall again be radiographed at a location rotated 90 degrees from the original view, with the spline nearest to the source. Projectiles and radiographs shall be suitably identified to permit cross reference. The lot represented by a sample with a defect shall be rejected.

5. PACKAGING.

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the

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managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The components and assemblies covered by this specification are intended to be used in the reassembly to Load, Assemble, and Pack M864E2 Projectiles utilizing self destruct fuzes staked to M42/M46 grenades.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Packaging requirements (see 5).
c. Requirements for submission of first article sample (see 3.1).
f. Government Furnished Equipment.
g. Requirements for submittal of Acceptance Inspection Equipment (see 6.3).
h. Requirements for ammunition lot numbering (in accordance with MIL-STD-1168).
i. Requirements for Certification of nondestructive test (NDT) personnel (see 6.4).
j. Requirements for submission of test data (see 6.5).

6.3 Submission of inspection equipment for designs approval. Submit equipment designs, as required, to Commander, ARDEC, ATTN: RDAR-QEM-A, Picatinny, NJ 07806-5000

6.4 Certification of nondestructive test (NDT) personnel. All personnel operating or calibrating NDT equipment should be qualified and certified in accordance with NAS-410. Qualification to other standards must be submitted to and approved by Commander, ARDEC, ATTN: RDAR-QEM-A, Picatinny, NJ 07806-5000.

6.5 Submission of test data. In addition to the normal distribution of records, when the M864 Projectiles are procured by the U.S. Army Tank Automotive Command, one (1) copy of all ballistic data and ammunition data cards should be forwarded to: Commander, U.S. Army ARDEC, ATTN: RDAR-QEM-A, Picatinny, NJ 07806-5000.

6.6 Visual examination qualification. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristics may be measured or gagged to determine acceptability.

6.7 Hybrid lotting by zone weight. Projectiles falling out of the inspection lot criteria due to the zone weight restriction may be accumulated into a hybrid lot.

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6.8 Alternative testing procedures. Alternative testing procedures may be used by the contractor when such testing and procedures provide as a minimum the quality assurance provisions specified herein. Prior to applying alternative procedures, the contractor is to describe them in a written proposal and should be prepared to assure that the effectiveness is equal to, or better than quality assurance provisions specified herein.

6.9 Drawings. Drawings listed in section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by and identified as U.S. Army Research and Development Command (ARRADCOM), Rock Island Arsenal, Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.10 Submission of alternative quality conformance provisions. All contractor- proposed alternative quality conformance provisions should be submitted to the Government for evaluation/approval by the technical activity responsible for the preparation of this specification.

6.11 Critical characteristic controls. All production contracts will contain requirements for handling critical defects. As part of all proposals, suppliers will be asked to describe policies, procedures and controls for all operations associated with safety characteristics, how they are documented and maintained under the supplier's integrated management system.

6.12 Inspection lot size. It is noted that the size of inspection lots of components, assemblies or item for delivery may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitate scheduling of tests or deliveries, inspection lots of items for delivery may be equivalent to such contract quantities provided all of the lot formation criteria and sampling provisions of this specification are maintained.

6.13 Ammunition lot numbers. Ammunition lot numbers requires ammunition data cards in accordance with MIL-STD-1168.

6.14 Definitions

6.14.1 Air Detonation: A grenade air detonation is defined as functioning of the grenade in the air due to any cause during the time interval after expulsion from the projectile and prior to impact on the prepared field.

6.14.2 Unexploded Ordinance (UXO): A UXO is defined as an item where any energetic material is still remaining in the M42 or M46 grenade, fuze assembly, or body loading assembly.

6.14.3 Premature burst: A premature burst is considered to have occurred if the projectile functions either in the gun tube or in flight prior to fuze set time.

6.14.4 Functioning (not including safety phase): Satisfactory functioning of the projectile (except safety phase), is defined as successful delivery of the cargo to the target area.

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6.14.5 Functioning (in the safety phase): Satisfactory functioning of the projectile, when fired at the safety phase, is defined as successful delivery of the cargo to the target area.

6.14.6 Critical Characteristics:

Critical characteristics are physical or functional attributes of a system, item, assembly, subassembly, component or material that, in the event of a nonconformance, is likely to:

1. Cause death, disability, injury or occupational illness that may result in hospitalization for individuals handling, using, storing, maintaining or depending upon the product;

2. Cause severe environmental damage that violates law or regulation;

3. Prevent performance of a tactical function of a system, exposing the user or allies to unacceptable levels of operational safety risk; or

4. Prevent performance of a tactical function of a weapon system or major item, where repair or replacement costs exceed $200,000 in Fiscal Year 2000 dollars.

6.14.7 Automatic Acceptance Inspection Equipment (AAIE): Use of non-operator-dependent, repeatable, automatic-decision-making inspection equipment.

6.14.8 Rotating band damaged. The machined rotating band should be free of nicks, burrs or otherwise damages subsequent from handling prior to shipment. Contractors should contact RDAR-QEM-A for damages in question.

6.15 Justification for critical classification

6.15.1 Distance from bottom of cover to tip of arming screw, max. The location of the Arming screw tip is essential to prevent the slider from arming prior to intended function. Should the slider arm during loading or assembly an initiation of the detonator would function the grenade causing injury or death.

6.15.2 Safety pin missing. The safety pin is one of two main safety features that is essential to maintain the slider in a safe position during manufacturing, shipping, and handling. Prior to loading the grenade into the projectile cavity, the slider on self destruct fuze is capable of moving to in-line firing position if the safety pin is missing. In the in-line firing position, the arming screw is then able to initiate the detonator. Safety pin also holds the trigger assembly in place, preventing the beam spring tip from striking the ignitor. This is critical because detonation in the armed position can cause injury or death.

6.15.3 Safety clip missing. The safety clip is one of two main safety features whose presence is essential to prevent disengagement of the arming screw with slider maintaining a safe position during manufacturing, shipping, and handling. Prior to loading the grenade into the projectile cavity, the slider on self destruct fuze is capable of moving to in-line firing position if the safety features are missing. If in the in-line firing position, the arming screw is able to

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initiate the detonator. This is critical because detonation in the armed position can cause injury or death

6.15.4 Slider protruding excessively from fuze indicating armed or partially armed condition. It is essential for the slider to remain in the fully safe position to prevent grenade detonation other than as intended. This is critical because unintentional functioning can cause injury or death

6.15.5 Slider Lock missing. The slider and tape loop are held in place by the slider lock. A missing slider lock could allow the slider to move to the armed position or cause the tape loop to not properly arm and function fuze. This is critical because unintentional functioning can cause injury or death.

6.15.6 Trigger missing or out of position. The beam spring is held in place by trigger assembly. A missing trigger could inadvertently cause the beam spring to strike the ignitor and set off the M55 detonator. A trigger out of position could inadvertently cause the beam spring to strike the ignitor and set off the pyrotechnic element and then the M55 detonator. The trigger is essential to prevent the ignitor from functioning during manufacturing, shipping, and handling. The beam spring restrained by the trigger maintains safety until deployment. This is critical because unintentional functioning can cause injury or death.

6.15.7 Layers 7 - 9 grenade type incorrect (yellow marking or not present). The M42 grenade body is not marked with the yellow band at its neck, and has serrations. If the M42 is used for layers 7-9, there can be an inadvertent initiation of grenade on setback due to the embossed body of the M42 grenade. The M46 grenade should be loaded in layers 7-9 of projectile. This is critical because unintentional functioning can cause injury or death.

6.15.8 Any grenade missing. The M864 projectile loads seventy-two (72) grenades within its cavity. A missing grenade could potentially cause an in-advertent initiation of grenade within gun tube during setback forces due to cavity within projectile. This is critical because unintentional functioning can cause injury or death.

6.15.9 Any sleeve, spacer, or spline missing. Sleeves, spacers, and splines provide a tight pack of grenades within the projectile body. A missing sleeve, spacer, or spline could cause a loose fit of grenades within the projectile body, causing the grenade cargo to move freely within projectile. On setback, grenade could possibly initiate. During flight, the trajectory of projectile could be off-set. This is critical because unintentional functioning can cause injury or death.

6.15.10 Adapter or adapter insert missing. Adapters and adapter inserts provide a tight pack of grenades within the projectile body. A missing adapter or adapter insert could cause a loose fit of grenades within the projectile body, causing the grenade to move back and forth. On setback, grenade could possibly initiate. This is critical because unintentional functioning can cause injury or death.

6.15.11 Excessive gap between projectile body and base burner. The projectile body threads could potentially shear, causing the projectile body and base burner to collide in gun tube

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on setback. Collision of projectile body and base burner could potentially cause metal parts break-up within gun tube and/or erratic projectile flight. A gap between projectile body and base burner could potentially allow propellant gases to enter the projectile cavity and initiate grenades. This is critical because unintentional functioning can cause injury or death.

6.15.12 Ballistic functioning safety. The projectile body should remain intact until initiation of artillery projectile fuze at the predetermined target area. This is critical because unintentional functioning prior projectile fuze being initiated can cause injury or death.

6.16 Radiographic maintenance. Radiographs should be maintained by the contractor until final acceptance of the projectile. If requirements of 3.6 are not met, the contractor should report findings to RDAR-QEM-A.

6.17 Safety warning. This projectile contains a pyrotechnic fuze that has a self-destruct/self-sterilization feature which will function regardless of the position of the slider. Personnel are strongly advised to be familiar with the safe handling of this fuze to prevent injury.

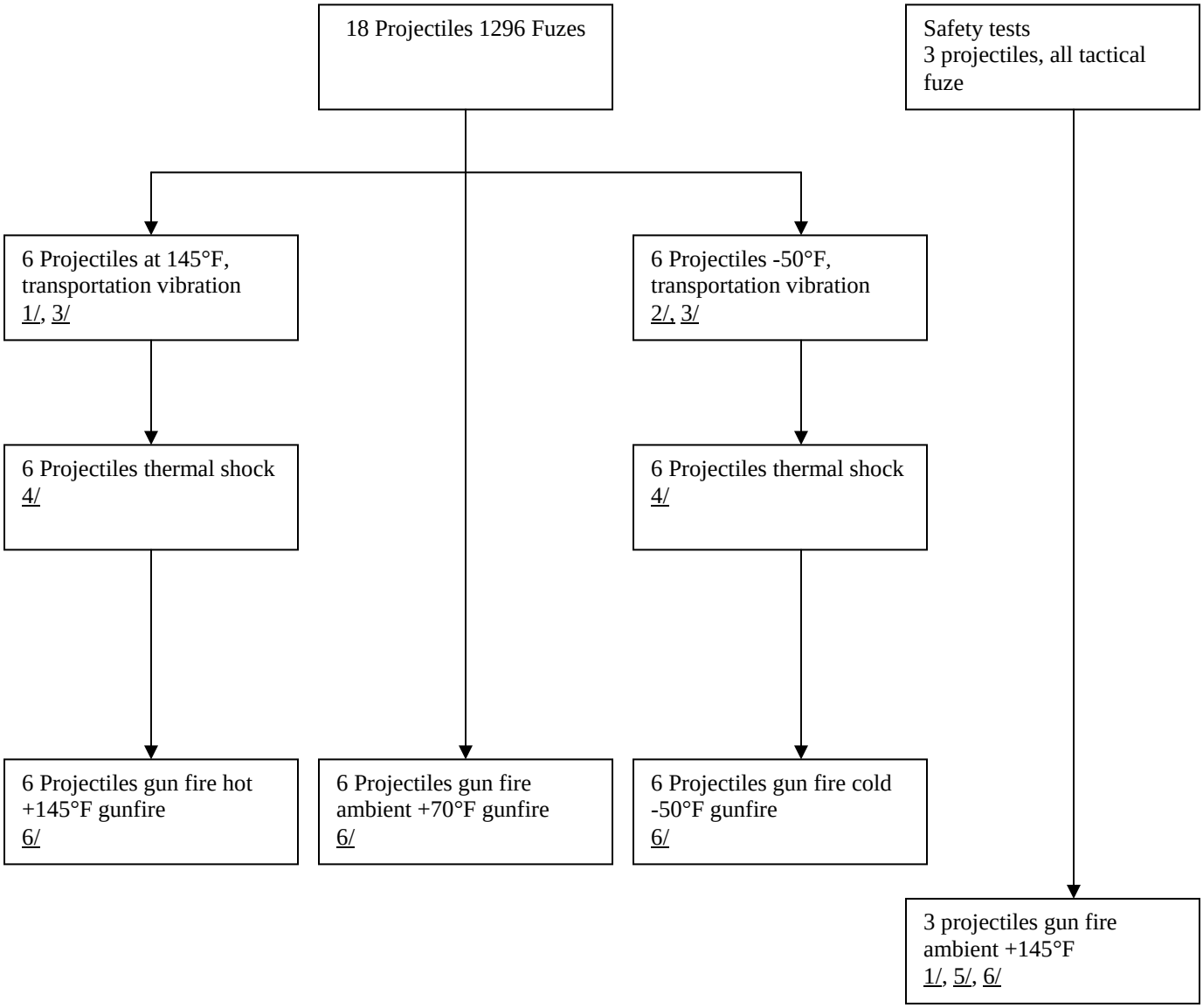
6.18 Figures. The below figures serve as an optional guide for the first article and lot acceptance tests.

6.19 Subject term (key word) listing.

Ammunition
Sub-munitions
Self-destruct

6.20 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

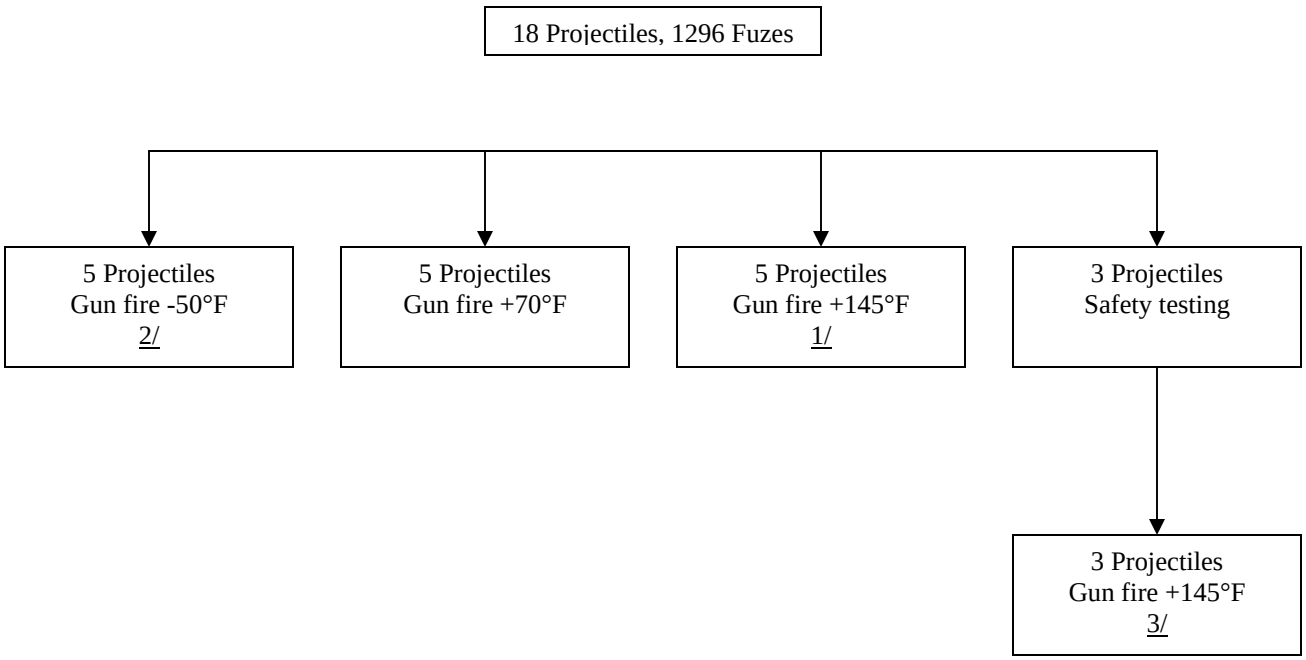
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NOTES:
1/ Condition full-up round and propellant hot +145°F, for a minimum of one hour.
2/ Condition full-up round and propellant cold -50°F, for a minimum of one hour.
3/ The sample shall be subject to transportation vibration in accordance with 4.4.10.6.
4/ The sample shall be subject to thermal shock in accordance with 4.4.10.5.
5/ Fire PIMP charge at approximately 63,000psi
6/ All grenades are live and fuzes tactical.

FIGURE 1. First article ballistic test

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NOTES:
1/ Condition full-up round and propellant hot +145°F, for a minimum of twenty four hours.
2/ Condition full-up round and propellant cold -50°F, for a minimum of twenty four hours.
3/ Condition round and propellant hot +145F fire pimp charge at approximately 60,300 psi.
4/ All grenades are live and fuzes tactical.

FIGURE 2. Lot acceptance ballistic test

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Preparing Activity:
Army – AR
(Project 1320-2010-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

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